

PhenotypeR

Review codelists and cohorts in OMOP CDM

2026-06-25

Introduction

- PhenotypeR package can help us to assess the research-readiness of a set of cohorts we have defined.
- The code is publicly available in OHDSI's GitHub repository **PhenotypeR**.
- PhenotypeR 0.5 is available in **CRAN**.
- Vignettes with further information can be found in the package **website**.

Set of Functions: Individual Diagnostics Assessment

- **Database diagnostics**
 - `databaseDiagnostics()`
- **Cohort diagnostics**
 - `cohortDiagnostics()`
- **Codelist diagnostics**
 - `codelistDiagnostics()`
- **Population diagnostics**
 - `populationDiagnostics()`

Set of Functions: Phenotype Diagnostics

- Comprises all the diagnostics that are being offered in this package.

```
result <- phenotypeDiagnostics(  
  cohort,  
  databaseDiagnostics = list(),  
  codelistDiagnostics = list(),  
  cohortDiagnostics = list(),  
  populationDiagnostics = list()  
)
```

- Run only some of the diagnostics:

```
result <- phenotypeDiagnostics(  
  cohort,  
  databaseDiagnostics = NULL,  
  codelistDiagnostics = list(),  
  cohortDiagnostics = NULL,  
  populationDiagnostics = list()  
)
```

Create a shiny app to visualise all the results!

- Create a shiny app to visualize all the results

```
shinyDiagnostics(result = result,  
                 directory = here(),  
                 minCellCount = 5,  
                 removeEmptyTabs = FALSE)
```

See an example in [here](#)

Database Diagnostics

- Summarise the database metadata including:
 1. **Snapshot**
 2. Summary of the **person table**
 3. Summary of the **observation period**
 4. Summary of the **clinical tables** (i.e., *condition_occurrence*) where the concepts of the codelist defining your cohort are found.

Database Diagnostics

```
db_diagnostics <- databaseDiagnostics(  
  cohort,  
  cohortId = NULL,  
  snapshot = TRUE,  
  personTableSummary = TRUE,  
  observationPeriodsSummary = TRUE,  
  clinicalRecordsSummary = FALSE  
)  
  
# Modify databaseDiagnostics in phenotypeDiagnostics:  
result <- phenotypeDiagnostics(  
  cohort,  
  databaseDiagnostics = list(  
    "cohortId" = c(1,2),  
    "snapshot" = FALSE  
  )  
)
```

Codelist Diagnostics

- Summarise the codelist associated with your cohort including:
 1. **Achilles codes use** (only if ACHILLES tables are present)
 2. **Orphan codes use** (only if ACHILLES tables are present)
 3. **Cohort code use**
 4. **Measurement code use** (only if measurement concepts are present in your codelist)
 5. **Drug diagnostics** (only if drug concepts are present in your codelist)

Codelist Diagnostics

```
cl_diagnostics <- codelistDiagnostics(  
  cohort,  
  cohortId = NULL,  
  achillesCodeUse = FALSE,  
  orphanCodeUse = TRUE,  
  cohortCodeUse = TRUE,  
  drugDiagnostics = FALSE,  
  drugDiagnosticsSample = 20000,  
  measurementDiagnostics = FALSE,  
  measurementDiagnosticsSample = 20000  
)  
  
# Modify codelistDiagnostics in phenotypeDiagnostics:  
result <- phenotypeDiagnostics(  
  cohort,  
  codelistDiagnostics = list(  
    "cohortId" = c(1,2),  
    "achillesCodeUse" = FALSE
```

Cohort Diagnostics

- Summarise your cohort characteristics:
 1. **Cohort count & attrition**
 2. **Cohort characteristics**: Baseline characteristics of your cohort
 3. **Large scale characteristics**: Summary of the records from clinical tables within a time window
 4. **Compare cohorts**: Overlap and timing between cohorts (only if more than one cohort are present)
 5. **Cohort survival**: Survival until the event of death (if death table is present)

Cohort Diagnostics

```
c_diagnostics <- cohortDiagnostics(  
  cohort,  
  cohortId = NULL,  
  cohortCount = TRUE,  
  cohortCharacteristics = TRUE,  
  largeScaleCharacteristics = TRUE,  
  compareCohorts = FALSE,  
  cohortSurvival = FALSE,  
  cohortSample = 20000,  
  matchedSample = 1000  
)  
  
# Modify cohortDiagnostics in phenotypeDiagnostics:  
result <- phenotypeDiagnostics(  
  cohort,  
  cohortDiagnostics = list(  
    "cohortSample" = 1000  
  ),  
)
```

Population Diagnostics

- Contextualises the frequency of your study cohorts in the database calculating:
 1. **Incidence**
 2. **Period Prevalence**

Population Diagnostics

```
c_diagnostics <- populationDiagnostics(  
  cohort,  
  cohortId = NULL,  
  incidence = TRUE,  
  periodPrevalence = TRUE,  
  populationSample = 1e+05,  
  populationDateRange = as.Date(c(NA, NA))  
)  
  
# Modify populationDiagnostics in phenotypeDiagnostics:  
result <- phenotypeDiagnostics(  
  cohort,  
  populationDiagnostics = list(  
    "populationSample" = 10000  
  )  
)
```

Summary:

- **Database diagnostics**
 - `databaseDiagnostics()`
- **Codelist diagnostics**
 - `codelistDiagnostics()`
- **Cohort diagnostics**
 - `cohortDiagnostics()`
- **Population diagnostics**
 - `populationDiagnostics()`
- Run all the diagnostics together:

```
result <- phenotypeDiagnostics(  
  cohort,  
  databaseDiagnostics = list(),  
  codelistDiagnostics = list(),  
  cohortDiagnostics = list(),  
  populationDiagnostics = list(),  
  stagingDirectory = NULL  
)  
  
shinyDiagnostics(result = result,  
  directory = here(),  
  minCellCount = 5,  
  removeEmptyTabs = FALSE)
```

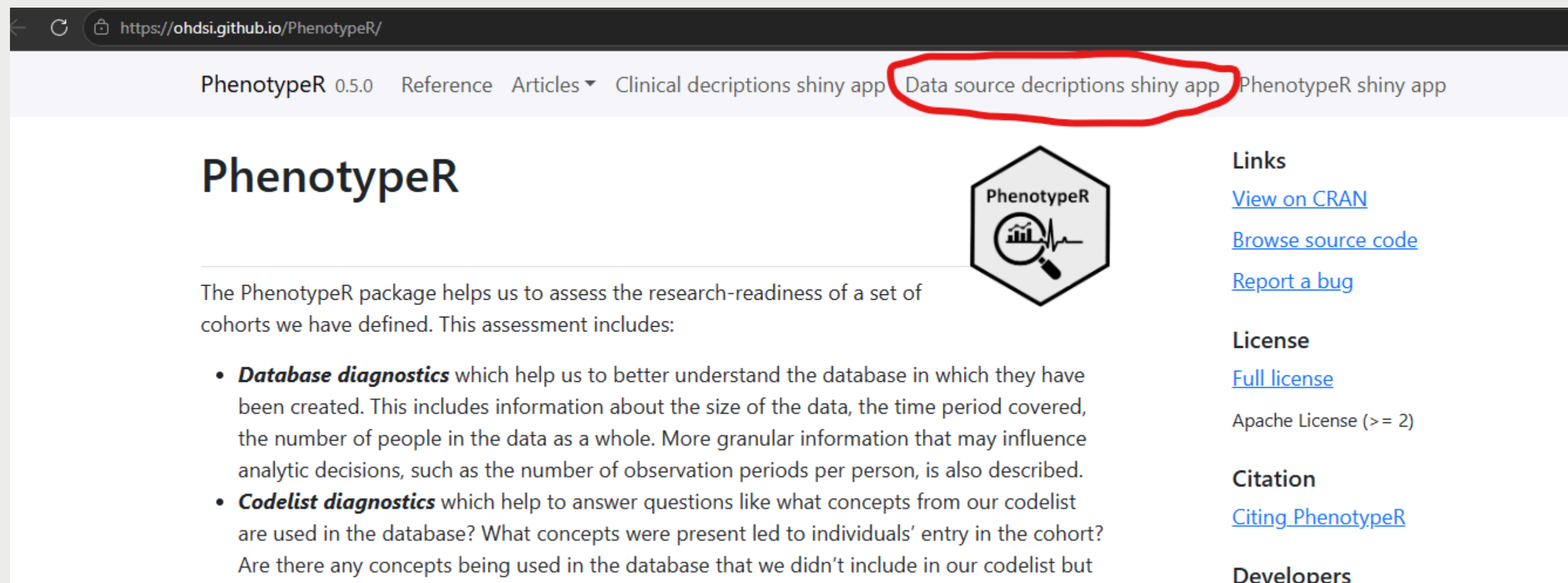

Extra: Changing additional parameters

- Modify extra parameters (*window*, *eventInWindow*, *episodesInWindow* for the LSC) using global variables:

```
options("PhenotypeR_summariseLargeScaleCharacteristics_window" = list(c(0, 0)))  
options("PhenotypeR_summariseLargeScaleCharacteristics_eventInWindow" = c("condition_occurrence"))  
options("PhenotypeR_summariseLargeScaleCharacteristics_episodeInWindow" = c("drug_exposure"))
```

Extra: Adding context to your results via data source description

- Visit [Data source description shiny app](#)



The screenshot shows the PhenotypeR website at <https://ohdsi.github.io/PhenotypeR/>. The navigation bar includes links for PhenotypeR 0.5.0, Reference, Articles, Clinical descriptions shiny app, **Data source descriptions shiny app** (highlighted with a red circle), and PhenotypeR shiny app. The main content area features the PhenotypeR logo, a description of the package's purpose, and a list of diagnostic categories. On the right, there are links for CRAN, source code, bug reports, license, citation, and developers.

PhenotypeR

The PhenotypeR package helps us to assess the research-readiness of a set of cohorts we have defined. This assessment includes:

- **Database diagnostics** which help us to better understand the database in which they have been created. This includes information about the size of the data, the time period covered, the number of people in the data as a whole. More granular information that may influence analytic decisions, such as the number of observation periods per person, is also described.
- **Codelist diagnostics** which help to answer questions like what concepts from our codelist are used in the database? What concepts were present led to individuals' entry in the cohort? Are there any concepts being used in the database that we didn't include in our codelist but

Links

- [View on CRAN](#)
- [Browse source code](#)
- [Report a bug](#)

License

[Full license](#)

Apache License (>= 2)

Citation

[Citing PhenotypeR](#)

Developers

Extra: Adding context to your results via data source description

- Visit [Data source description shiny app](#)

Data source description

A specification for data source descriptions

Administrative details

Name of data source ⓘ

Data source acronym ⓘ

Data source website ⓘ

HMA-EMA catalogue entry ⓘ

Main references ⓘ

Data collected

Database Description

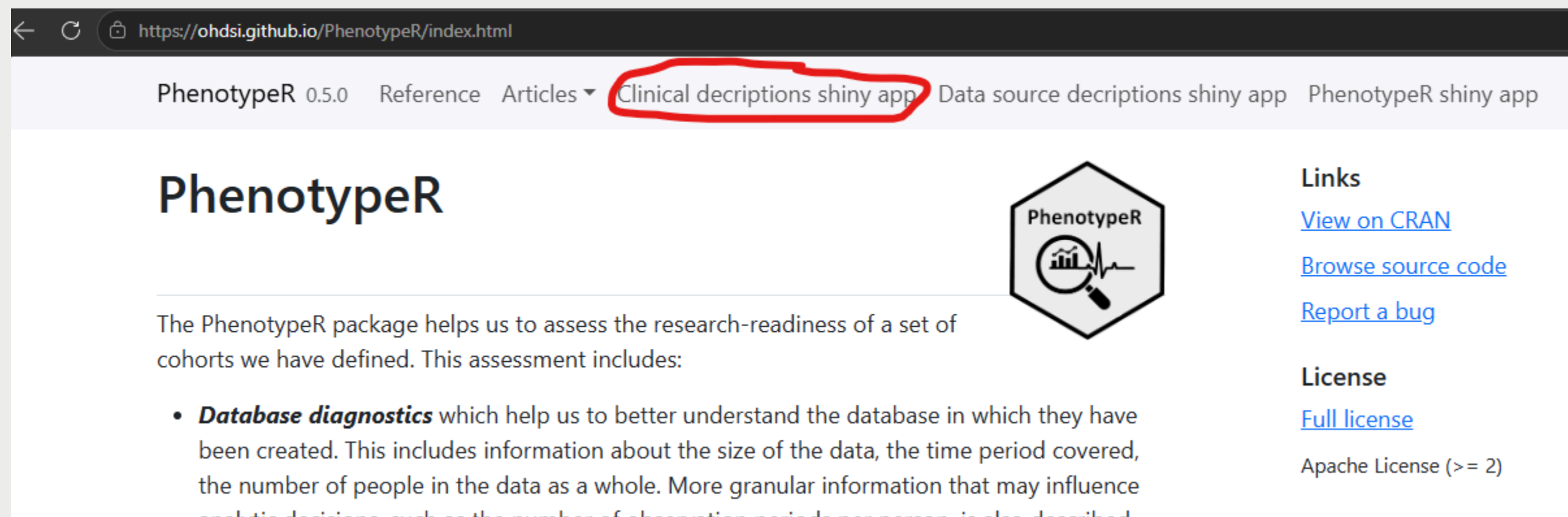
Geography ⓘ

Real World Evidence Summer School 2026

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Extra: Adding context to your results via clinical descriptions

- Visit [Clinical Descriptions shiny app](#)



The screenshot shows the PhenotypeR website at <https://ohdsi.github.io/PhenotypeR/index.html>. The navigation bar includes links for PhenotypeR 0.5.0, Reference, Articles, Clinical descriptions shiny app (highlighted with a red circle), Data source descriptions shiny app, and PhenotypeR shiny app. The main content area features the title "PhenotypeR" and a description: "The PhenotypeR package helps us to assess the research-readiness of a set of cohorts we have defined. This assessment includes:" followed by a bullet point: "• **Database diagnostics** which help us to better understand the database in which they have been created. This includes information about the size of the data, the time period covered, the number of people in the data as a whole. More granular information that may influence analytic decisions, such as the number of observation periods per person, is also described." To the right of the text is a hexagonal logo with the word "PhenotypeR" and a magnifying glass over a bar chart. Further right is a "Links" section with links to "View on CRAN", "Browse source code", and "Report a bug", and a "License" section with a link to "Full license" and the text "Apache License (>= 2)".

Extra: Adding context to your results via clinical descriptions

- Visit [Clinical descriptions shiny app](#)

Clinical Description

A clinical description to inform development of computable phenotypes using real world health data.

🔒 Phenotype name

Phenotype Name ⓘ

✂️ DRAFT WITH AI

🔒 Metadata

Version ⓘ

Created By ⓘ

Created Date ⓘ

2026-06-18

Last Edited By ⓘ

Last Edited Date ⓘ

2026-06-18

Source of Information ⓘ

🔒 Clinical Profile

Introduction & Synonyms ⓘ

Import your descriptions in the shiny app:

```
shinyDiagnostics(result = result,  
  directory = here(),  
  minCellCount = 5,  
  clinicalDescriptionsDir = NULL,  
  databaseDescriptionsDir = NULL,  
  removeEmptyTabs = FALSE)
```


PhenotypeR

👉 [Package website](#)

👉 [CRAN link](#)

👉 [GitHub link](#)

👉 [Manual](#) 

